SAMPLE PRE-ACTION DIGITAL EVIDENCE TRIAGE WORKPAPER (100% SYNTHETIC)

Matter Reference: SYN-PILOT-2025-0100 (100% Clean-Room Specification Scenario — Zero Real Client Data)
Scenario Type: UK Micro-Enterprise Domestic Faster Payments Supplier Invoice Diversion (`PSR Specific Direction 20` Scope)
Prepared By: Ziqi Xiao (`starshield-forensics v0.3.7` Deterministic Triage Workflow · OpenSSF Silver `#16602` · Scorecard `8.0/10` · OSPS Baseline L3)
Engagement Basis: Pre-Action Technical Consultancy Workpaper for Instructing Solicitors (Fixed-Fee Tier, Non-Contingent; Not a CPR Part 35 Report)
Document Classification: TLP:CLEAR — Reproducible Synthetic Sample Deliverable

1. Executive Summary & Scope of Instructions

1. **Fictional Fact Pattern:** On 18 March 2025 (post-`7 October 2024` implementation of mandatory APP reimbursement under **PSR Specific Direction 20, para 3.2**), a fictional UK micro-enterprise ("Example-Co Precision Synthetic Ltd", 6 employees, annual turnover £620k — qualifying as a protected micro-enterprise) remitted **GBP 18,400.00 via UK Faster Payments** (within the £85,000 PS24/7 regulatory maximum reimbursement level) after receiving amended domestic settlement details inside a browser/Electron-based procurement portal thread impersonating its regular UK component vendor. Ten minutes after composition, the impersonator soft-deleted the payment instruction (`"This message was deleted"` in UI screenshots) and sent a follow-up reserve cross-border IBAN/card demand (`msg_spec_0002`), which was halted unpaid.
2. **Scope of Technical Triage:** Instructed on a synthetic demonstration basis to: (a) inspect a consented, read-only copy of the Chromium / Electron IndexedDB write-ahead log (`000003.log`) and exported JSON archive (`synthetic_case_benchmark.json`) to recover soft-deleted records prior to LevelDB compaction; (b) validate extracted banking identifiers via deterministic ISO/IEC 7812 Luhn, ISO 13616 IBAN Mod-97, and ISO 9362 BIC rules; (c) inspect RFC 2083 PNG container metadata on a disputed "Stamped Bank Audit Notice"; and (d) compile an offset-cited schedule aligned with `UK FSR Statutory Code (FSA — DIG 100)` & `ISO/IEC 17025` for solicitor review.

2. Reproducible Chain-of-Custody & Read-Only Acquisition Manifest

(Every byte length, `SHA-256` digest, and byte offset below matches the deterministic `starshield-forensics v0.3.7` specification vectors 1-to-1.)

Exhibit ID	Synthetic Specification Vector	Exact Byte Length	Reproducible SHA-256 Digest	Verification Module
EXH-01	`build_synthetic_leveldb_wal_bytes()` (`000003.log`)	68,779 B	`c2ac19e91535c2a641023ac96893c2d5c25a38c3e5f507254e91e75bf6b172ff`	`leveldb_wal_32k_crc32c_v0.3.6`
EXH-02	`samples/synthetic_case_benchmark.json`	2,570 B	`a2a722157007057ee1eb9de5a01b89c1c5080a36fc9ab433c97387aa3321e7b2`	`triage_exported_chatsvc_json`
EXH-03	`build_valid_png_with_iccp("Display P3", "APPL", ...)`	195 B	`068e9a47ee0f1444b22b04bd151a9c5cae57b90ab8f43e4a1c256c2106aea2bf`	`inspect_image_container (RFC 2083)`

> **Provenance Caveat:** Cryptographic `SHA-256` digests attest post-acquisition byte integrity only. Upstream workstation custody and absence of prior LevelDB compaction depend on the client intake log recorded at acquisition.

3. Recovered Soft-Deleted Cache Records & Checksum-Validated Indicators

Parsing `EXH-01` across three `32,768`-byte LevelDB blocks (`FULL` + `FIRST / MIDDLE / LAST` reassembly; `4` valid physical frames; `0` Castagnoli `CRC32C` errors) recovered two soft-deleted WriteBatch JSON records hidden in UI views by `properties.deletetime`:

Seq #	Physical WAL Byte Offset	Message ID	Compose UTC / Soft-Delete UTC	Verbatim Cached UTF-8 Payload (`OBSERVED`)	Checksum-Validated IOCs (`DERIVED`)	Epistemic Tier

1001	0x00000000	msg_spec_0001	2025-03-18 10:00:00Z Deleted: 2025-03-18 10:10:00Z	"Updated UK Faster Payments supplier account for Invoice INV-2025-0100 (GBP 18,400.00): Sort Code 00-00-00 Account 00000000."	UK Sort Code 00-00-00 paired within 120-char window to 8-digit account 00000000 (test vector)	OBSERVED (Bytes) DERIVED (Pairing)
1002	0x0000018D (3-block span)	msg_spec_0002	2025-03-18 14:30:00Z Deleted: 2025-03-18 14:40:00Z	"Secondary reserve IBAN GB82WEST12345698765432 (SWIFT NWBKGB2LXXX), Norway test IBAN NO9386011117947, and Luhn test card 6200000000000005."	• IBANs GB82... (len 22) & NO93... (len 15): **Mod-97 PASS** • BIC NWBKGB2LXXX: **ISO-3166 PASS** • Card 6200...0005: **Luhn PASS**	OBSERVED (Bytes) DERIVED (Mod-97 / Luhn)

4. Disputed Container Provenance & PSR Specific Direction 20 Schedule

4.1 Structural PNG Container Inspection (EXH-03 — remittance_stamp_2025.png)

- **Observed Structure (OBSERVED / DERIVED):** Valid 8-byte RFC 2083 PNG magic, IHDR (1 x 1 minimal spec vector at offset 8), iCCP chunk at byte offset 33 (CRC32 valid) decompressing to a 160-byte ISO 15076-1 acsp profile (Primary Platform: APPL , Description: Display P3), tEXt chunk at byte offset 111 (Software\x00QuickShot Raster Editor 4.2), IDAT , and terminal IEND .
- **Mandatory Alternative Explanations:** Apple Display P3 profiles and raster editor tEXt tags show consumer/raster pipeline export rather than native banking host PDF generation. Because staff occasionally screenshot legitimate PDFs on macOS/iOS, this is reported strictly as a **corroborative provenance discrepancy (DERIVED)**, never standalone proof of forgery.

4.2 Chronological Alignment for UK Faster Payments Reimbursement (PSR Specific Direction 20)

Under **PSR Specific Direction 20 and Pay.UK FPS Reimbursement Rules** (applicable to eligible UK Faster Payments executed on or after 7 October 2024 by consumers, micro-enterprises, or small charities with annual income <£1m to a **"recipient account not controlled by the consumer"**), the schedule isolates objective evidence addressing PSP "Consumer Standard of Caution" objections:

UTC Timestamp (2025-03-18)	Source Exhibit & Offset	Objective Technical Observation	Relevance to Solicitor's PSR SD20 / FOS Submission
10:00:00Z	EXH-01 (offset 0x00000000)	msg_spec_0001 delivers UK Sort Code 00-00-00 / 00000000 inside existing supplier thread.	Reconstructs contemporaneous inducement text deleted from UI screenshots prior to claim intake.
10:00:45Z	EXH-02 (timestamp_pairs[0])	Disputed PNG audit notice (EXH-03 , raster tEXt offset 111) acknowledged (+45.0s UTC delta).	Corroborates active social-engineering artefact presented before Faster Payments authorization.
10:06:18Z	Claimant Bank Statement	Outbound UK Faster Payment (GBP 18,400.00) debited to third-party UK recipient account.	Post- 7 Oct 2024 UK Faster Payment within **PSR SD20** scope; recipient account not controlled by claimant.
14:30:00Z	EXH-01 (offset 0x0000018D)	Unpaid follow-up msg_spec_0002 adds cross-border NO93... IBAN & card rail; retracted at 14:40:00Z .	Cross-border SEPA/card rails fall outside FPS SD20 reimbursement scope, while corroborating third-party fraudster control.

5. Quality Assurance Log & Methodological Boundaries

1. **Cache Compaction Boundary:** LevelDB background compaction purges overwritten entries; absence of earlier deleted records in EXH-01 does not prove no prior retractions occurred.
2. **Sort-Code Weighting:** Validates Luhn, ISO-13616 Mod-97 (89 country lengths), and ISO-9362 BICs deterministically; UK Sort Code pairing uses a 120-char intra-message window without proprietary Vocalink modulus tables.
3. **Offline Execution, Dual-Tool Validation & QA Sign-Off:** Stdlib-only Python >= 3.10 (zero network calls , zero LLM uploads , single-file starshield.pyz & uvx --from starshield-forensics starshield demo included); regression suite **25/25 passed**; 482-vector dual-tool validation aligned with **UK FSR Statutory Code of Practice (FSA — DIG 100 , superseding FSR-C-107) & ISO/IEC 17025** ; continuous Atheris/ClusterFuzzLite fuzzing (OpenSSF Scorecard 8.0/10); all digests and offsets verified directly against starshield-forensics v0.3.7 (DOI: 10.5281/zenodo.24156262 · SWHID: swh:1:snp:4ed5efecab5e40975e75a5c5107b63121b1df60a).